DAU Cooling Fan – General data

1. General

Air is circulated around the Data Acquisition Units by a three port cooling fan installed just aft of the DAUs.

2. Description

A Sandia ACF 328 cooling fan is installed in the ceiling to provide air circulation for the DAUs. Power to the fan is controlled by the thermal switch which is located on the DAU maintenance panel around station 244.00. The thermal switch is set to activate when the temperature reaches 110 degrees Fahrenheit (43 degrees Celsius).

The DAU cooling fan is mounted in the aircraft ceiling around station 270.00. Hoses run from the fan ports, one hose runs between the DAUs, the other two hoses run on either side of the DAUs.

28 Volt DC power is provided from the left bus through a 3 amp DAU COOLING FAN (U2) circuit breaker on station 332.00 circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).

3. Adjustment/Test

A. DAU Cooling Fan Test

1) While monitoring temperature, apply heat to the thermal switch.

2) The switch should close at a temperature of 110 ± 8 degrees Fahrenheit (43 degrees Celsius). Confirm fan is operating.

3) While monitoring temperature, allow the thermal switch to cool.

4) The switch should open at a temperature of 100 ± 5 degrees Fahrenheit. Confirm fan is no longer running.

B. Thermal Switch Adjustment

1) Remove thermal switch if required.

2) Rotate set screw until the pointer reads 110 degrees Fahrenheit.

4. Removal and Installation

A. DAU Cooling Fan

1) Removal of DAU Cooling Fan

a) Ensure the aircraft electrical system is OFF.

b) Open the DAU COOLING FAN (U2) circuit breaker on the station 332.00 circuit breaker panel and fit safety clip.

c) Gain access to the ceiling compartment aft of bulkhead 262.

d) Disconnect electrical connector.

e) Loosen (3) hose clamps and remove hoses from fan ports.

f) Remove the (9) mounting screws.

g) Remove unit from the bracket.

Note Stow connector of removed components.

2) Installation of DAU Cooling Fan

a) Ensure the aircraft electrical system is OFF.

b) Open the DAU COOLING FAN (U2) circuit breaker on the station 332.00 circuit breaker panel and fit safety clip.

c) Set unit in position and install (9) mounting screws.

d) Install hoses to fan ports and tighten (3) hose clamps.

e) Reconnect electrical connector. Ensure connection is secure.

f) Install interior as required.

g) Remove safety clip and close DAU COOLING FAN (U2) circuit breaker on the station 332.00 circuit breaker panel.

B. Thermal Switch

1) Removal of Thermal Switch

a) Ensure the aircraft electrical system is OFF.

b) Open the DAU COOLING FAN (U2) circuit breaker on the station 332.00 circuit breaker panel and fit safety clip.

c) Gain access to the ceiling compartment aft of bulkhead 240.

d) Remove (2) electrical terminals. Protect terminals from short circuits to airframe or each other.

e) Remove (2) mounting screws.

f) Remove switch from panel.

2) Installation of Thermal Switch

a) Ensure the aircraft electrical system is OFF.

b) Ensure switch is set to the proper temperature, refer to Para 3.B.

c) Set switch in position and install (2) mounting screws.

d) Remove protection and install (2) electrical terminals. Ensure terminals are tight.

e) Install interior as required.

f) Remove safety clip and close DAU COOLING FAN (U2) circuit breaker on the station 332.00 circuit breaker panel.

5. Troubleshooting

A. DAU Cooling Fan

1) Ensure DAU COOLING FAN (U2) circuit breaker is closed.

2) Visually inspect the connector for damaged wiring or contacts. Ensure electrical connector is fully seated.

3) Ensure thermal switch is operating normally, refer to Para 4.B.

4) If troubleshooting at a wire level is required, refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

5) If repair or service is required, send to appropriately rated avionics repair facility.

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